

Linear Regression

STAT 429: Introduction to Time Series Analysis

Sabyasachi Chatterjee

Fall 2026

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1 Simple Linear Regression

We shall start the first topic of the course: Linear Regression. We start our discussion with simple linear regression (where there is a single covariate), and then extend to multiple linear regression (where there are multiple covariates).

We want to learn the relationship between two variables y and x , with the aim of predicting y given the value of x . y is called the response variable, and x is called the covariate. For example (this was one of the original applications of regression), y denotes the height of an adult man, and x denotes the height of their father. The linear regression model assumes that y is related to x via the equation:

$$y = \beta_0 + \beta_1 x + \epsilon \quad (1)$$

where β_0 and β_1 are parameters, and ϵ denotes an error term which captures deviations of y from the assumed equation $\beta_0 + \beta_1 x$. The parameters β_0 and β_1 can be interpreted as follows: β_0 denotes the value of y when $x = 0$ and β_1 represents the change in y when x changes by one unit.

We observe data $(x_1, y_1), \dots, (x_n, y_n)$ on the covariate and response variables corresponding to n instances (in the height example, we have data on heights for n father-son pairs). Writing the equation (1) for each individual pair (x_i, y_i) we get

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (2)$$

The observed data $(x_1, y_1), \dots, (x_n, y_n)$ will be used to obtain estimates $\hat{\beta}_0$ and $\hat{\beta}_1$ (as well as uncertainty quantification) for the parameters β_0 and β_1 . After obtaining these estimates, one can predict the value of the response variable for a possibly new covariate value x_{new} by $\hat{\beta}_0 + \hat{\beta}_1 x_{\text{new}}$.

We will implement the equations for obtaining $\hat{\beta}_0$ and $\hat{\beta}_1$ from the observed data $(x_1, y_1), \dots, (x_n, y_n)$ using the Python library `statsmodels`. We will also study the math behind this process.

To apply linear regression, we need data on both y and x . In the time series context, the observed data is $y_1, \dots, y_n$ which represent observations for a single variable y . There is no additional data on another variable x . In order to apply regression methods to time series, we need to create a covariate variable x . There are two main ways of doing it:

1. **Time as covariate:** Here we take the time index as the covariate x . For example, in the time series dataset on the population of the United States for each month from January 1959 to December 2024: n denotes the total number of data points, $x_i = i$ and y_i denotes the observed population data for the i^{th} month (first month is January 1959, second month is February 1959 and so on).
2. **Lagged y as covariate:** Here we take $x_i = y_{i-1}$. In other words, the covariate equals the response at the previous time point. This kind of regression is called Lagged Regression or, more commonly, AutoRegression.

For now, we shall focus on the first kind of regression (time as covariate). We shall study AutoRegression in more detail later.

1.1 Estimation of β_0 and β_1

1.1.1 Least Squares Estimates

The estimates of β_0 and β_1 reported by standard libraries (such as `statsmodels`) are obtained using the method of least squares. This involves minimizing the sum of squares criterion:

$$S(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \quad (3)$$

over all values of β_0 and β_1 . It is left as an exercise to verify that:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad \text{and} \quad \hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2},$$

where

$$\bar{y} = \frac{y_1 + \cdots + y_n}{n} \quad \text{and} \quad \bar{x} = \frac{x_1 + \cdots + x_n}{n}.$$

1.1.2 MLE under Normality of Errors

Suppose we assume that the error terms $\epsilon_1, \dots, \epsilon_n$ in (2) are i.i.d normal with mean zero and some variance σ^2 :

$$\epsilon_1, \dots, \epsilon_n \stackrel{\text{i.i.d}}{\sim} N(0, \sigma^2). \quad (4)$$

Then the least squares estimates of β_0 and β_1 coincide with the Maximum Likelihood Estimates (MLEs). We use the function `OLS` in the Python library `statsmodels` to do inference on β_0 and β_1 (inference refers to estimating their values, and also to obtain uncertainty intervals).

There are two main approaches for statistical inference: Frequentist and Bayesian. We shall explore how each of these work for linear regression.

2 Frequentist Inference

The basic ideas behind frequentist inference are as follows:

1. Construct a method for estimating the unknown parameters. In the simple linear regression context, one can use least squares to estimate β_0 and β_1 . Probably the most popular estimation strategy is Maximum Likelihood Estimation (this requires writing down a likelihood function).
2. Calculate (exactly or using some approximations) the distribution of the estimators. Use quantiles of the distribution for obtaining interval estimates for the unknown parameters. The quantiles might themselves depend on other unknown parameters (which would then have to be replaced by estimates).

To minimize $S(\beta_0, \beta_1)$, we take derivatives with respect to β_0, β_1 and equate to zero:

$$\begin{aligned} \frac{\partial}{\partial \beta_0} S(\beta_0, \beta_1) = 0 &\implies \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i) = 0 \\ \frac{\partial}{\partial \beta_1} S(\beta_0, \beta_1) = 0 &\implies \sum_{i=1}^n x_i (y_i - \beta_0 - \beta_1 x_i) = 0 \end{aligned} \quad (5)$$

Maximum Likelihood Estimation (MLE) can also be used. With the normality assumption (4), the linear regression model can be rewritten as:

$$y_i \stackrel{\text{independent}}{\sim} N(\beta_0 + \beta_1 x_i, \sigma^2). \quad (6)$$

The likelihood then becomes:

$$\begin{aligned} f_{y_1, \dots, y_n | \beta_0, \beta_1, \sigma}(y_1, \dots, y_n) &= \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2}\right) \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2\right) \\ &= (2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) \end{aligned} \quad (7)$$

Recall $S(\beta_0, \beta_1)$ above is the sum of squares defined in (3). To write this likelihood, we are assuming that $x_1, \dots, x_n$ are fixed. This assumption is fine if $x_i = i$ (regression with time as covariate) but not strictly true when $x_i = y_{i-1}$ (auto-regression). We shall see how it is still approximately true in the case of AutoRegression later.

The Maximum Likelihood Estimates of the parameters β_0, β_1, σ are obtained by maximizing the likelihood. As maximizing a function is equivalent to maximizing its logarithm, we attempt to maximize the log-likelihood which leads to an easier maximization. The log-likelihood is:

$$\begin{aligned} \text{log-likelihood} &= \log \left[(2\pi)^{-n/2} \sigma^{-n} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) \right] \\ &= -\frac{n}{2} \log(2\pi) - n \log \sigma - \frac{S(\beta_0, \beta_1)}{2\sigma^2}. \end{aligned}$$

To maximize the log-likelihood, we simply take derivatives with respect to the unknown parameters β_0, β_1, σ and equate those to zero:

$$\begin{aligned} \frac{\partial}{\partial \beta_0} \text{log-likelihood} &= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta_0} S(\beta_0, \beta_1) = 0 \implies \frac{\partial}{\partial \beta_0} S(\beta_0, \beta_1) = 0 \\ \frac{\partial}{\partial \beta_1} \text{log-likelihood} &= -\frac{1}{2\sigma^2} \frac{\partial}{\partial \beta_1} S(\beta_0, \beta_1) = 0 \implies \frac{\partial}{\partial \beta_1} S(\beta_0, \beta_1) = 0 \\ \frac{\partial}{\partial \sigma} \text{log-likelihood} &= -\frac{n}{\sigma} + \frac{S(\beta_0, \beta_1)}{\sigma^3} = 0 \implies \sigma = \sqrt{\frac{S(\beta_0, \beta_1)}{n}}. \end{aligned}$$

The first two equations coincide with the corresponding equations (5) for minimizing least squares. This shows that the MLEs for β_0 and β_1 coincide with the least squares estimators (19). The MLE for σ is given by the third equation above (with β_0 and β_1 replaced by $\hat{\beta}_0$ and $\hat{\beta}_1$ respectively):

$$\hat{\sigma}_{\text{MLE}} = \sqrt{\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n}}. \quad (8)$$

The next step is to determine the distribution of the estimators. Let us illustrate this with $\hat{\beta}_1$:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (y_i - \bar{y})(x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

where the second equality uses $\sum_{i=1}^n \bar{y}(x_i - \bar{x}) = \bar{y} \sum_{i=1}^n (x_i - \bar{x}) = 0$. Under the normality assumption (6), it is easy to check that

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n y_i (x_i - \bar{x})}{\sum_{i=1}^n (x_i - \bar{x})^2} \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}\right).$$

For $\hat{\beta}_0$, one can derive a similar normal distribution (proof omitted):

$$\hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n \sum_{i=1}^n (x_i - \bar{x})^2}\right). \quad (9)$$

For $\hat{\sigma}_{\text{MLE}}^2$, the distribution becomes (proof omitted):

$$\frac{n\hat{\sigma}_{\text{MLE}}^2}{\sigma^2} \sim \chi_{n-2}^2 \quad (10)$$

where χ_{n-2}^2 is the chi-squared distribution with $n - 2$ degrees of freedom. The mean of the chi-squared distribution equals its degrees of freedom which implies that

$$\mathbb{E}\hat{\sigma}_{\text{MLE}}^2 = \sigma^2 \frac{n-2}{n}.$$

Therefore the MLE for σ^2 is not unbiased (in contrast, the MLEs $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased). It is easy to correct the bias leading to the following unbiased estimator of σ^2 :

$$\hat{\sigma}_{\text{unbiased}}^2 = \frac{n}{n-2} \hat{\sigma}_{\text{MLE}}^2 = \frac{S(\hat{\beta}_0, \hat{\beta}_1)}{n-2}.$$

Usage of $\hat{\sigma}_{\text{unbiased}}$ is much more common than that of $\hat{\sigma}_{\text{MLE}}$ (note that $\hat{\sigma}_{\text{unbiased}}$ is not unbiased for σ ; rather the square of $\hat{\sigma}_{\text{unbiased}}$ is unbiased for σ^2).

Another important fact is that $(\hat{\beta}_0, \hat{\beta}_1)$ and $\hat{\sigma}_{\text{unbiased}}^2$ are independent.

These facts are used to derive the following confidence interval for β_1 :

$$\left[\hat{\beta}_1 - \frac{\hat{\sigma}_{\text{unbiased}}}{\sqrt{\sum_i (x_i - \bar{x})^2}} t_{n-2, \alpha/2}, \hat{\beta}_1 + \frac{\hat{\sigma}_{\text{unbiased}}}{\sqrt{\sum_i (x_i - \bar{x})^2}} t_{n-2, \alpha/2} \right] \quad (11)$$

where $t_{n-2, \alpha/2}$ is the positive point such that $\mathbb{P}\{t_{n-2} \geq t_{n-2, \alpha/2}\} = \alpha/2$ (i.e., the t -distribution with $n - 2$ degrees of freedom assigns probability mass $\alpha/2$ to the right of $t_{n-2, \alpha/2}$). (11) is a valid confidence interval because:

$$\frac{\hat{\beta}_1 - \beta_1}{\sigma_{\text{unbiased}}} \sqrt{\sum_i (x_i - \bar{x})^2} \sim N(0, 1) \text{ and } \frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}_{\text{unbiased}}} \sqrt{\sum_i (x_i - \bar{x})^2} \sim t_{n-2}$$

where t_{n-2} is the t -distribution with $n - 2$ degrees of freedom.

3 A First Introduction to Bayesian Inference

Suppose a coin is tossed n times and we observe m heads. Let p denote the unknown bias of the coin, so that, conditional on p , each toss results in a head with probability p . We would like to answer questions such as

$$\Pr\{p \in [a, b] \mid M = m\},$$

where M denotes the number of heads observed in the n tosses.

Thomas Bayes's crucial step was to regard the unknown bias itself as a random variable. Philosophically, this is the essential Bayesian move: probability is used not only to describe the

randomness in the data, but also to represent our uncertainty about an unknown parameter. Suppose, for example, that

$$p \sim \text{Unif}(0, 1).$$

This is our **prior distribution** for p .

Conditional on p , the number of heads has the binomial distribution

$$\Pr\{M = m \mid p\} = \binom{n}{m} p^m (1 - p)^{n-m}.$$

This conditional probability, viewed as a function of p for the observed value $M = m$, is the **likelihood**.

Bayes's rule now gives

$$\begin{aligned} \Pr\{p \in [a, b] \mid M = m\} &= \frac{\Pr\{p \in [a, b], M = m\}}{\Pr\{M = m\}} \\ &= \frac{\int_a^b \binom{n}{m} p^m (1 - p)^{n-m} dp}{\Pr\{M = m\}}. \end{aligned} \quad (12)$$

The denominator can be found using Bayes's billiard-table argument. The first ball lands at a uniformly chosen point $p \in [0, 1]$. Conditional on this point, each of the next n balls falls to one side of the first ball with probability p . Among the $n + 1$ possible ranks of the first ball, each rank is equally likely. Consequently, the number of the remaining balls falling to that side is uniformly distributed on $\{0, 1, \dots, n\}$. Therefore,

$$\Pr\{M = m\} = \frac{1}{n + 1}.$$

Substituting this into (12), we obtain

$$\Pr\{p \in [a, b] \mid M = m\} = (n + 1) \binom{n}{m} \int_a^b p^m (1 - p)^{n-m} dp.$$

The integral in this expression must then be evaluated. Equivalently, because

$$(n + 1) \binom{n}{m} = \frac{1}{B(m + 1, n - m + 1)},$$

we can write

$$\Pr\{p \in [a, b] \mid M = m\} = \frac{\int_a^b p^m (1 - p)^{n-m} dp}{\int_0^1 p^m (1 - p)^{n-m} dp}.$$

Thus, after observing m heads in n tosses, the **posterior distribution** of the bias is

$$p \mid M = m \sim \text{Beta}(m + 1, n - m + 1).$$

This example contains the general logic of Bayesian inference. In any statistical model involving unknown parameters, we proceed as follows:

1. Assign a probability distribution to the unknown parameters. This is the **prior distribution**.
2. Specify the conditional distribution of the observed data given the parameters. For the observed data, this gives the **likelihood**.
3. Apply Bayes's rule to obtain the conditional distribution of the parameters given the observed data. This is the **posterior distribution**:

$$\text{posterior} \propto \text{likelihood} \times \text{prior}.$$

4. Use the posterior distribution to answer inferential questions about the unknown parameters.

4 Bayesian Inference for Simple Linear Regression

We use the prior

$$\beta_0, \beta_1, \log \sigma \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(-C, C)$$

for a large number C (the exact value of C will not matter in the following calculations). Note that as σ is always positive, we have made the uniform assumption on $\log \sigma$ (by the change of variable formula, the density of σ would be given by $f_\sigma(x) = f_{\log \sigma}(\log x) \frac{1}{x} = \frac{I\{-C < \log x < C\}}{2Cx} = \frac{I\{e^{-C} < x < e^C\}}{2Cx}$).

The joint posterior for all the unknown parameters β_0, β_1, σ is then given by (below we write the term “data” for $y_1, \dots, y_n$):

$$f_{\beta_0, \beta_1, \sigma | \text{data}}(\beta_0, \beta_1, \sigma) \propto f_{y_1, \dots, y_n | \beta_0, \beta_1, \sigma}(y_1, \dots, y_n) f_{\beta_0, \beta_1, \sigma}(\beta_0, \beta_1, \sigma).$$

The two terms on the right hand side above are the likelihood:

$$f_{y_1, \dots, y_n | \beta_0, \beta_1, \sigma}(y_1, \dots, y_n) \propto \sigma^{-n} \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2 \right) = \sigma^{-n} \exp \left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2} \right),$$

and the prior:

$$\begin{aligned} f_{\beta_0, \beta_1, \sigma}(\beta_0, \beta_1, \sigma) &= f_{\beta_0}(\beta_0) f_{\beta_1}(\beta_1) f_\sigma(\sigma) \\ &\propto \frac{I\{-C < \beta_0 < C\}}{2C} \frac{I\{-C < \beta_1 < C\}}{2C} \frac{I\{e^{-C} < \sigma < e^C\}}{2C\sigma} \\ &\propto \frac{1}{\sigma} I\{-C < \beta_0, \beta_1, \log \sigma < C\}. \end{aligned}$$

Recall that $S(\beta_0, \beta_1)$ denotes the sum of squares:

$$S(\beta_0, \beta_1) := \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2.$$

We thus obtain

$$f_{\beta_0, \beta_1, \sigma | \text{data}}(\beta_0, \beta_1, \sigma) \propto \sigma^{-n-1} \exp \left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2} \right) I\{-C < \beta_0, \beta_1, \log \sigma < C\}.$$

The above is the joint posterior over β_0, β_1, σ . The posterior over only the main parameters β_0, β_1 can be obtained by integrating (or marginalizing) the parameter σ .

$$\begin{aligned} f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) &= \int f_{\beta_0, \beta_1, \sigma | \text{data}}(\beta_0, \beta_1, \sigma) d\sigma \\ &\propto I\{-C < \beta_0, \beta_1 < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) d\sigma. \end{aligned}$$

When C is large, the above integral can be evaluated from 0 to ∞ which gives

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto I\{-C < \beta_0, \beta_1 < C\} \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) d\sigma.$$

The change of variable

$$s = \frac{\sigma}{\sqrt{S(\beta_0, \beta_1)}}$$

allows us to write the integral as

$$\begin{aligned} &\int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \beta_1)}{2\sigma^2}\right) d\sigma \\ &= \left(\frac{1}{S(\beta_0, \beta_1)}\right)^{n/2} \int_0^\infty s^{-n-1} \exp\left(-\frac{1}{2s^2}\right) ds \\ &\propto \left(\frac{1}{S(\beta_0, \beta_1)}\right)^{n/2}. \end{aligned}$$

where, we are using the fact that $\int_0^\infty s^{-n-1} \exp(-1/(2s^2)) ds$ is a constant (in the sense that it does not depend on β_0 and β_1).

The posterior density of (β_0, β_1) is thus

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto I\{-C < \beta_0, \beta_1 < C\} \left(\frac{1}{S(\beta_0, \beta_1)}\right)^{n/2}. \quad (13)$$

In words, the posterior density is inversely proportional to $S(\beta_0, \beta_1)^{n/2}$. This implies that the posterior mode is just the least squares estimator $(\hat{\beta}_0, \hat{\beta}_1)$. It is nicer to write the posterior in the following equivalent form:

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)}\right)^{n/2} I\{-C < \beta_0, \beta_1 < C\} \quad (14)$$

Note that (13) and (14) represent exactly the same density because the term $(S(\hat{\beta}_0, \hat{\beta}_1))^{n/2}$ does not depend on β_0, β_1 and is thus a constant.

Generally, the density (14) will be quite sharply concentrated around the least squares estimator $(\hat{\beta}_0, \hat{\beta}_1)$ especially when n is large. This is because, when (β_0, β_1) is such that $S(\beta_0, \beta_1)$ is large compared to $S(\hat{\beta}_0, \hat{\beta}_1)$, the quantity

$$\left(\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)}\right)^{n/2}$$

would be quite negligible because of the large power $n/2$. As a result, the posterior density $f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1)$ will be concentrated around those values of (β_0, β_1) for which $S(\beta_0, \beta_1)$

is quite close to $S(\hat{\beta}_0, \hat{\beta}_1)$. For example, suppose $n = 791$, and that (β_0, β_1) is such that $S(\beta_0, \beta_1) = (1.1)S(\hat{\beta}_0, \hat{\beta}_1)$. Then

$$\left(\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)} \right)^{n/2} = \left(\frac{1}{1.1} \right)^{395.5} \approx 4.26 \times 10^{-17}.$$

Such (β_0, β_1) will thus get negligible posterior probability. Even for (β_0, β_1) such that $S(\beta_0, \beta_1) = (1.01)S(\hat{\beta}_0, \hat{\beta}_1)$, we have

$$\left(\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)} \right)^{n/2} = \left(\frac{1}{1.01} \right)^{395.5} \approx 0.02$$

and so such (β_0, β_1) will also get fairly small posterior probability.

To sum up, when n is large, the posterior probability will be concentrated around those (β_0, β_1) for which $S(\beta_0, \beta_1)$ is very close to $S(\hat{\beta}_0, \hat{\beta}_1)$. Generally, this would imply that (β_0, β_1) would itself have to be close to $(\hat{\beta}_0, \hat{\beta}_1)$. For this reason, the indicator term in (14) has no effect when C is large. From now on, we shall drop this indicator term and refer to the Bayesian posterior as simply

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1)}{S(\beta_0, \beta_1)} \right)^{n/2}. \quad (15)$$

It turns out that this represents a multivariate t -density, as we shall soon. Before that, let us first note that we have essentially the same formula in multiple linear regression as well.

5 Multiple Linear Regression

In multiple linear regression, we have one response variable y and m covariates $x_1, \dots, x_m$ ($m = 1$ corresponds to simple linear regression). We observe data on n instances or subjects for all these variables: $(y_i, x_{i1}, \dots, x_{im})$ for $i = 1, \dots, n$. The multiple linear regression model (with normal errors) is given by:

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_m x_{im} + \epsilon_i \quad \text{with } \epsilon_i \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2). \quad (16)$$

In the time series context, the model (16) arises in the following ways:

1. **Regression with functions of time:** Suppose we want to fit a quadratic function of time to the data. We can do this via the model (16) with x_1 being the time variable, and x_2 being the squared time i.e., $x_{i1} = i$ and $x_{i2} = i^2$. Suppose we want to fit a simple sinusoidal function to the data. We can do this via (16) with $x_{i1} = \cos(2\pi i/12)$ and $x_{i2} = \sin(2\pi i/12)$.
2. **AutoRegression (AR):** If we take $x_{ij} = y_{i-j}$, then we get the AR model:

$$y_t = \beta_0 + \beta_1 y_{t-1} + \dots + \beta_m y_{t-m} + \epsilon_t.$$

The idea here is that we are using the m most recent values of the time series to predict the next observation. AR models are very commonly used and they work quite well for time series.

In Bayesian inference for (16), we work with the prior

$$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \stackrel{\text{i.i.d}}{\sim} \text{unif}(-C, C)$$

for a very large positive C . The joint posterior density of $\beta_0, \dots, \beta_m, \sigma$ is then given by

$$\begin{aligned} & f_{\beta_0, \beta_1, \dots, \beta_m, \sigma | \text{data}}(\beta_0, \beta_1, \dots, \beta_m, \sigma) \\ & \propto \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) I\{-C < \beta_0, \beta_1, \dots, \beta_m, \log \sigma < C\}. \end{aligned}$$

where we use the notation

$$S(\beta_0, \beta_1, \dots, \beta_m) := \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_{i1} - \dots - \beta_m x_{im})^2$$

for the sum of squares.

The posterior over only the coefficient parameters β_0, β_1 can be obtained by integrating (or marginalizing) the parameter σ .

$$\begin{aligned} & f_{\beta_0, \beta_1, \dots, \beta_m | \text{data}}(\beta_0, \beta_1, \dots, \beta_m) \\ &= \int f_{\beta_0, \beta_1, \dots, \beta_m, \sigma | \text{data}}(\beta_0, \beta_1, \dots, \beta_m, \sigma) d\sigma \\ &\propto I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \int_{e^{-C}}^{e^C} \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) d\sigma \\ &\approx I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \int_0^\infty \sigma^{-n-1} \exp\left(-\frac{S(\beta_0, \dots, \beta_m)}{2\sigma^2}\right) d\sigma \\ &= I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \int_0^\infty s^{-n-1} \exp\left(-\frac{1}{2s^2}\right) ds \\ &\propto I\{-C < \beta_0, \beta_1, \dots, \beta_m < C\} \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \\ &\approx \left(\frac{1}{S(\beta_0, \dots, \beta_m)}\right)^{n/2} \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)}{S(\beta_0, \beta_1, \dots, \beta_m)}\right)^{n/2} \end{aligned}$$

where $\hat{\beta}_0, \dots, \hat{\beta}_m$ denote the least squares estimators of $\beta_0, \dots, \beta_m$ (i.e., $(\hat{\beta}_0, \dots, \hat{\beta}_m)$ minimizes $S(\beta_0, \dots, \beta_m)$ over all values of $\beta_0, \dots, \beta_m$).

Our posterior density for $\beta_0, \dots, \beta_m$ is thus:

$$f_{\beta_0, \beta_1, \dots, \beta_m | \text{data}}(\beta_0, \beta_1, \dots, \beta_m) \propto \left(\frac{S(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_m)}{S(\beta_0, \beta_1, \dots, \beta_m)}\right)^{n/2}. \quad (17)$$

Now we will explain why this is a multivariate t -density.

5.1 Why is (17) a t -density?

If you go to the wikipedia page (https://en.wikipedia.org/wiki/Multivariate_t-distribution) for Multivariate t -distribution, it gives the following formula for the density:

$$\begin{aligned} f(x) &:= \frac{\Gamma((\nu + p)/2)}{\Gamma(\nu/2) \nu^{p/2} \pi^{p/2} \sqrt{\det \Sigma}} \left[\frac{1}{1 + \frac{1}{\nu}(x - \mu)^T \Sigma^{-1}(x - \mu)} \right]^{(\nu+p)/2} \\ &\propto \left[\frac{1}{1 + \frac{1}{\nu}(x - \mu)^T \Sigma^{-1}(x - \mu)} \right]^{(\nu+p)/2}. \end{aligned} \quad (18)$$

Their notation for this distribution is $t_p(\mu, \Sigma, \nu)$ where:

1. p denotes dimension of the vector x (this is a p -variate joint density)
2. μ is a $p \times 1$ vector called the location
3. Σ is a $p \times p$ matrix called the scale matrix
4. $\nu > 0$ denotes the degrees of freedom.

It turns out that (17) is a special case of (18) for some p, μ, Σ, ν . To see this, we need to first rewrite (17) using matrix notation which we do in the next section.

5.2 Matrix Notation for Multiple Linear Regression

$$y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \quad X = \begin{pmatrix} 1 & x_{11} & \dots & x_{1m} \\ 1 & x_{21} & \dots & x_{2m} \\ \vdots & \vdots & \dots & \vdots \\ 1 & x_{n1} & \dots & x_{nm} \end{pmatrix} \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix} \quad \hat{\beta} = \begin{pmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_m \end{pmatrix}$$

This notation is used not just to write formulae for linear regression, but also in code. For example, the OLS function in `statsmodels` uses the syntax `sm.OLS(y, X).fit()` to fit the linear regression model, where y ($n \times 1$ vector) and X ($n \times (m+1)$ matrix) are defined above.

With this notation, one can write the sum of squares $S(\beta_0, \dots, \beta_m)$ as:

$$S(\beta) = S(\beta_0, \dots, \beta_m) = \|y - X\beta\|^2.$$

There are two important facts about $S(\beta)$:

1. **Fact 1:** the least squares estimator $\hat{\beta}$ is given by the formula:

$$\hat{\beta} = (X^T X)^{-1} X^T y. \quad (19)$$

The proof of (19) is as follows. The gradient of $S(\beta)$ is given by

$$\begin{aligned} \nabla S(\beta) &= \nabla [\|y - X\beta\|^2] \\ &= \nabla [(y - X\beta)^T (y - X\beta)] \\ &= \nabla [y^T y - \beta^T X^T y - y^T X \beta + \beta^T X^T X \beta] = 2X^T y - 2X^T X \beta. \end{aligned}$$

Because $\hat{\beta}$ minimizes $S(\beta)$, the gradient should equal zero when $\beta = \hat{\beta}$, and this leads to

$$X^T (y - X\hat{\beta}) = 0 \implies X^T X \hat{\beta} = X^T y \implies \hat{\beta} = (X^T X)^{-1} X^T y. \quad (20)$$

2. **Fact 2:** The following Pythagorean identity holds:

$$S(\beta) = S(\hat{\beta}) + \|X\beta - X\hat{\beta}\|^2 = S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta}). \quad (21)$$

To prove (21), write

$$\begin{aligned} S(\beta) &= \|y - X\beta\|^2 \\ &= \|y - X\hat{\beta} + X\hat{\beta} - X\beta\|^2 \\ &= \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2 + 2\langle y - X\hat{\beta}, X\hat{\beta} - X\beta \rangle. \end{aligned}$$

The cross product is zero (leading to (21)) because:

$$\begin{aligned} \langle y - X\hat{\beta}, X\hat{\beta} - X\beta \rangle &= (X\hat{\beta} - X\beta)^T (y - X\hat{\beta}) \\ &= (\hat{\beta} - \beta)^T X^T (y - X\hat{\beta}) = (\hat{\beta} - \beta)^T (X^T y - X^T X\hat{\beta}) = 0 \end{aligned}$$

where we used (20).

Using (21), we can write the posterior density (14) as

$$\begin{aligned} f_{\beta|\text{data}}(\beta) &\propto \left(\frac{S(\hat{\beta})}{S(\beta)} \right)^{n/2} \\ &= \left(\frac{S(\hat{\beta})}{S(\hat{\beta}) + (\beta - \hat{\beta})^T X^T X (\beta - \hat{\beta})} \right)^{n/2} \\ &= \left(\frac{1}{1 + (\beta - \hat{\beta})^T \frac{X^T X}{S(\hat{\beta})} (\beta - \hat{\beta})} \right)^{n/2}. \end{aligned} \quad (22)$$

The above formula is a special case of (18) with

$$x = \beta, \quad p = m + 1, \quad \mu = \hat{\beta}, \quad \nu + p = n, \quad \frac{\Sigma^{-1}}{\nu} = \frac{X^T X}{S(\hat{\beta})}$$

or equivalently

$$x = \beta, \quad p = m + 1, \quad \mu = \hat{\beta}, \quad \nu = n - m - 1, \quad \Sigma = \frac{S(\hat{\beta})}{n - m - 1} (X^T X)^{-1}.$$

We thus have

$$\beta_0, \dots, \beta_m \mid \text{data} \sim t_{m+1} \left(\hat{\beta}, \frac{S(\hat{\beta})}{n - m - 1} (X^T X)^{-1}, n - m - 1 \right). \quad (23)$$

With the posterior density (23), one can do uncertainty quantification about the parameters $\beta_0, \beta_1, \dots, \beta_m$. One can generate multiple samples from $t_{m+1}(\hat{\beta}, (S(\hat{\beta})/(n - m - 1))(X^T X)^{-1}, n - m - 1)$ and plot the resulting fitted values to visualize the uncertainty in the coefficients.

5.3 Posterior t -density in Multiple Linear Regression

In (23), $S(\hat{\beta})$ denotes the sum of squares evaluated at the least squares estimator. It is the smallest possible value of the sum of squares, and it is also known as the Residual Sum of Squares (RSS).

(23) represents the joint density of $(\beta_0, \dots, \beta_m)$ given the observed data. It turns out that the posterior distribution of each individual β_j is also given by a t -density. This follows from properties of the multivariate t -density which we go over next.

6 The Multivariate t -Distribution

Here is some more information about the t -density (18):

1. **Connection to the Multivariate Normal Density:** The most important term in the formula (18) is $(x - \mu)^T \Sigma^{-1} (x - \mu)$. This exact term also appears in the multivariate normal density. If $X \sim N(\mu, \Sigma)$, then the density of X is given by:

$$\frac{1}{(2\pi)^{p/2} \sqrt{\det \Sigma}} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

This suggests that the t -density is closely related to the multivariate normal density. Here is the connection. Suppose $X \sim N_p(\mu, \Sigma)$ and $V \sim \chi_\nu^2$ (this is the chi-squared distribution with ν degrees of freedom) are independent. Then

$$T := \mu + \frac{X - \mu}{\sqrt{V/\nu}} \sim t_p(\mu, \Sigma, \nu). \quad (24)$$

Thus, in the notation $t_p(\mu, \Sigma, \nu)$, ν denotes degrees of freedom, p denotes dimension, μ and Σ denote the mean vector and covariance matrix of the corresponding normal random vector X . For completeness, we include a proof of (24) in Section 8.

2. **Individual Components as well as Linear Combinations of Components of T are also t -distributed:** Suppose $T \sim t_p(\mu, \Sigma, \nu)$ and the components of T are $T_1, \dots, T_p$. Then each individual component T_j is also t -distributed. Also every linear combination $a_0 + a_1 T_1 + a_2 T_2 + \dots + a_p T_p$ is also t -distributed. To see this, first write

$$a_0 + a_1 T_1 + \dots + a_p T_p = a_0 + a^T T$$

where a is the $p \times 1$ vector with components $a_1, \dots, a_p$. Using the formula (24), we can write

$$a_0 + a^T T = (a_0 + a^T \mu) + \frac{(a_0 + a^T X) - (a_0 + a^T \mu)}{\sqrt{V/\nu}}$$

Because $a_0 + a^T X \sim N(a_0 + a^T \mu, a^T \Sigma a)$, the same fact (24) applied to this case gives:

$$a_0 + a^T T \sim t_1(a_0 + a^T \mu, a^T \Sigma a, \nu).$$

In particular, this implies that for each $j = 1, \dots, p$,

$$T_j \sim t_1(\mu_j, \Sigma(j, j), \nu)$$

where μ_j is the j th component of μ and $\Sigma(j, j)$ is the (j, j) th entry of Σ .

3. **When ν is large, t is very close to normal:** This can intuitively be seen by noting that when ν is large, the term $(x - \mu)^T \Sigma^{-1} (x - \mu) / \nu$ is small so that

$$1 + \frac{1}{\nu} (x - \mu)^T \Sigma^{-1} (x - \mu) \approx \exp \left(\frac{1}{\nu} (x - \mu)^T \Sigma^{-1} (x - \mu) \right),$$

where we used the observation that $1 + z \approx e^z$ when z is small. Thus the t -density (18) for large ν becomes approximately:

$$\exp \left(-\frac{\nu + p}{2\nu} (x - \mu)^T \Sigma^{-1} (x - \mu) \right) \approx \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right)$$

because $\frac{\nu+p}{\nu} \approx 1$ when ν is large. This gets us the normal density:

$$t_p(\mu, \Sigma, \nu) \approx N_p(\mu, \Sigma) \quad \text{if } \nu \text{ is large.}$$

In our regression case, the degrees of freedom is $n - m - 1$ where n is the number of observations, and m is the number of covariates. Thus **if $n - m - 1$ is large**, then the posterior distribution (which is actually t) is approximately normal:

$$t_{m+1}\left(\hat{\beta}, \frac{S(\hat{\beta})}{n - m - 1} (X^T X)^{-1}, n - m - 1\right) \approx N_{m+1}\left(\hat{\beta}, \frac{S(\hat{\beta})}{n - m - 1} (X^T X)^{-1}\right).$$

7 Back to Regression

The quantity $S(\hat{\beta})/(n - m - 1)$ is the **frequentist unbiased** estimator for σ^2 , so we denote it by $\hat{\sigma}^2$:

$$\hat{\sigma} := \sqrt{\frac{S(\hat{\beta})}{n - m - 1}}.$$

$\hat{\sigma}$ can also be justified as a Bayesian estimator of σ (See Question 5 (e) of Homework One). The terminology **Residual Standard Error** is sometimes used for $\hat{\sigma}$.

With the notation for $\hat{\sigma}$, the posterior (23) becomes:

$$\beta_0, \dots, \beta_m \mid \text{data} \sim t_{m+1}\left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}, n - m - 1\right). \quad (25)$$

By one of the facts mentioned about the t -distribution, the posterior of each individual β_j is also t :

$$\beta_j \mid \text{data} \sim t_1\left(\hat{\beta}_j, \hat{\sigma}^2 (X^T X)^{j+1, j+1}, n - m - 1\right) \quad (26)$$

where $(X^T X)^{j+1, j+1}$ is the $(j + 1)^{\text{th}}$ diagonal entry of $(X^T X)^{-1}$ (note that we are using the $(j + 1)$ th diagonal entry of $X^T X$ because β_j is the $(j + 1)$ th component of β). Writing this density out, we have

$$f_{\beta_j \mid \text{data}}(\beta_j) \propto \left(\frac{1}{1 + \frac{1}{n - m - 1} \frac{(\beta_j - \hat{\beta}_j)^2}{\hat{\sigma}^2 (X^T X)^{j+1, j+1}}} \right)^{n/2}$$

which implies that

$$\frac{\beta_j - \hat{\beta}_j}{\hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}}} \sim \text{univariate standard } t \text{ with } n - m - 1 \text{ d.f.}$$

This can be used to obtain uncertainty intervals for β_j . If $t_{n-m-1, \alpha/2}$ is the point beyond which the t -distribution (with $n - m - 1$ degrees of freedom) assigns probability $\alpha/2$, then

$$\mathbb{P} \left\{ -t_{n-m-1, \alpha/2} \leq \frac{\beta_j - \hat{\beta}_j}{\hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}}} \leq t_{n-m-1, \alpha/2} \mid \text{data} \right\} = 1 - \alpha$$

which is same as:

$$\mathbb{P} \left\{ \hat{\beta}_j - \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2} \leq \beta_j \leq \hat{\beta}_j + \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2} \mid \text{data} \right\} = 1 - \alpha$$

This interval:

$$\left[\hat{\beta}_j - \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2}, \hat{\beta}_j + \hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}} t_{n-m-1, \alpha/2} \right]$$

is called the $100(1 - \alpha)\%$ Bayesian Credible interval for β_j . It **exactly coincides** with the frequentist $100(1 - \alpha)\%$ confidence interval for β_j .

When $n - m - 1$ is large, the t -density (25) is approximately equal to the $N_{m+1}(\hat{\beta}, \hat{\sigma}^2(X^T X)^{-1})$. Further, when $n - m - 1$ is large, the distribution (26) will be close to the normal distribution $N(\hat{\beta}_j, \hat{\sigma}^2(X^T X)^{j+1, j+1})$. The quantity $\hat{\sigma} \sqrt{(X^T X)^{j+1, j+1}}$ is known as the standard error corresponding to β_j .

8 Proof of (24)

Proof of (24). Start with the formula:

$$f_T(y) = \int_0^\infty f_{T|V=x}(y) f_V(x) dx.$$

Observe that

$$T \mid V = x \sim N\left(\mu, \frac{\nu}{x} \Sigma\right)$$

so that

$$\begin{aligned} f_{T|V=x}(y) &= \frac{1}{(2\pi)^{p/2} \sqrt{\det(\frac{\nu}{x} \Sigma)}} \exp \left[-\frac{1}{2} (y - \mu)^T \left(\frac{\nu}{x} \Sigma \right)^{-1} (y - \mu) \right] \\ &= \frac{x^{p/2}}{(2\pi)^{p/2} \nu^{p/2} \sqrt{\det(\Sigma)}} \exp \left(-\frac{x}{2\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right) \end{aligned}$$

where we used $\det(\frac{\nu}{x} \Sigma) = (\nu/x)^p \det(\Sigma)$. As a result

$$\begin{aligned} f_T(y) &= \int_0^\infty f_{T|V=x}(y) f_V(x) dx \\ &\propto \int_0^\infty \frac{x^{p/2}}{(2\pi)^{p/2} \nu^{p/2} \sqrt{\det(\Sigma)}} \exp \left(-\frac{x}{2\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right) x^{\frac{\nu}{2}-1} e^{-x/2} dx \\ &\propto \int_0^\infty x^{\frac{p+\nu}{2}-1} \exp \left(-\frac{x}{2} \left[1 + \frac{1}{\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right] \right) dx. \end{aligned}$$

The change of variable

$$t = x \left[1 + \frac{1}{\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right]$$

leads to

$$\begin{aligned} f_T(y) &\propto \frac{1}{\left[1 + \frac{1}{\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right]^{\frac{\nu+p}{2}}} \int_0^\infty t^{\frac{\nu+p}{2}-1} e^{-t/2} dt \\ &\propto \frac{1}{\left[1 + \frac{1}{\nu} (y - \mu)^T \Sigma^{-1} (y - \mu) \right]^{\frac{\nu+p}{2}}}. \end{aligned}$$

which proves (24). □